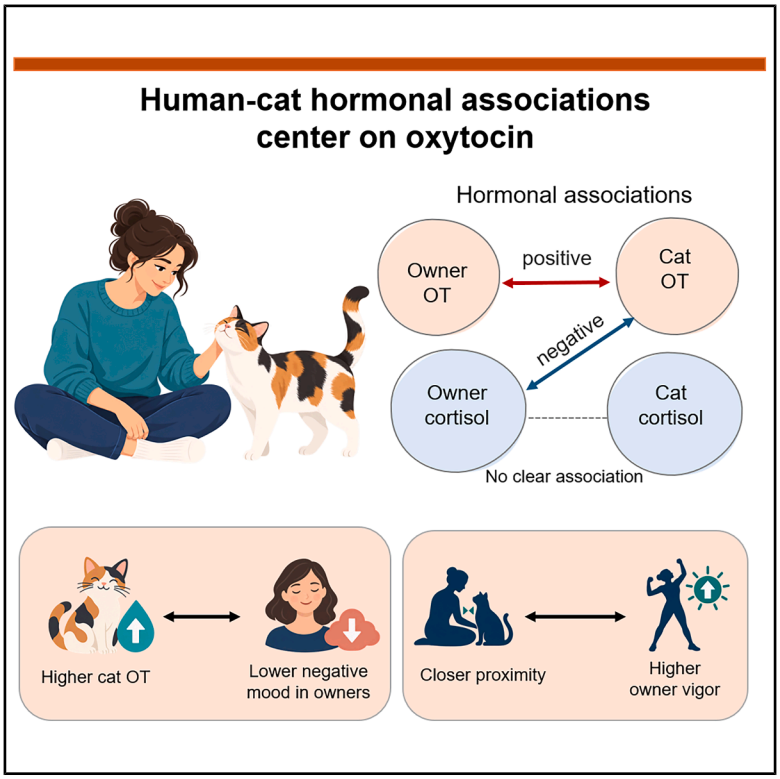
iScience

Article

Cats and humans show correlated affiliative physiological states

Graphical abstract



Highlights

- Owner and cat urinary oxytocin concentrations were positively correlated
- Higher cat oxytocin linked to lower owner cortisol and negative mood
- Owner-cat proximity, not cohabitation duration, predicted owner vigor
- Cat oxytocin and cortisol relate to distinct social approach behaviors

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In brief

Feline behavior; Biological sciences; Animal physiology

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Cats and humans show correlated affiliative physiological states

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SUMMARY

Cats are among the world's most popular companion animals. Owner-pet hormonal synchrony is documented in dogs but untested in cats. We measured oxytocin and cortisol in 19 owners and 29 pet cats over one month, and examined owner mood and owner-cat proximity via Bluetooth beacons. Owner and cat oxytocin were positively associated, whereas their cortisol concentrations showed little association. Higher cat oxytocin was linked to lower owner cortisol and less negative mood, while higher owner cortisol was linked to more negative mood. Closer proximity predicted higher owner vigor, and exploratory behavioral testing linked cat oxytocin to social approach and cortisol to reduced approach toward strangers. Unlike the cortisol-centered synchrony reported in human-dog dyads, human-cat relationships may center on oxytocin, extending interspecies hormonal findings to a species with a distinct domestication history. These correlations may partly reflect shared household environments rather than dyad-specific interactions, an issue that future studies should address.

INTRODUCTION

Humans and domestic cats (*Felis catus*) have coexisted for at least 2,000 years, and cats are now one of the most popular companion animals worldwide.[1,2] In Japan alone, approximately 8,847,000 cats are kept as pets, surpassing the dog population of 6,820,000, with this gap continuing to widen.[3] While pet ownership broadly confers health benefits—including reduced illness and improved psychological well-being—the effects of cat ownership appear to differ substantially from those of dog ownership. Bao and Schreer found that dog owners scored higher than cat owners across all dimensions of well-being, as well as on measures of personality traits, emotion regulation strategies, and need satisfaction.[4] In a 10-month prospective study, Serpell tracked behavioral and health changes in adults who newly acquired a dog or cat[5]; health benefits persisted for the full 10 months in dog owners, whereas benefits in cat owners were significant only during the first month and did not differ from controls thereafter. Similarly, Endo et al. reported that dog ownership was positively associated with mental well-being, while cat ownership showed a negative association.[6] Taken together, the effects of cat ownership are inconsistent and frequently diverge from those of dog ownership. These discrepancies may reflect differences in owner husbandry practices and the nature of daily human-cat interactions, yet the mechanisms by which such interactions influence the physiological and psychological state of cat owners remain poorly understood.

One candidate mechanism involves the hypothalamic-pituitary-adrenal (HPA) axis, a key hormonal network that regulates cortisol secretion in response to stress.[7] Oxytocin (OT) modulates HPA-axis activity by attenuating cortisol responses,[8] thereby exerting anti-stress effects[9] that may contribute to reduced cardiovascular disease risk.[10,11] These hormonal markers are also closely linked to psychological states, including emotion.[12] Such bidirectional physiological and psychological interactions may extend to human-animal relationships, and have been documented most extensively in the context of human-dog dyads.

Physiological synchrony between humans and dogs is well established. Nagasawa et al. demonstrated that mutual gazing between owners and dogs activates an oxytocin positive-feedback loop analogous to that observed in human parent-infant bonding.[13] Neural synchrony has also been reported, with electroencephalographic signals found to synchronize between humans and dogs during mutual eye contact.[14] At the hormonal level, long-term cortisol concentrations have been shown to correlate between dogs and their owners.[15]

Whether analogous synchrony exists between humans and cats remains an open question. Examining hormonal measures in both members of owner–cat dyads may reveal whether everyday affiliative interactions are associated with the

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physiological states of both partners, rather than with those of either the owner or the cat alone. It may also clarify whether cross-species physiological associations are a general feature of companion-animal relationships or differ according to species-specific patterns of domestication and social interaction.

Regarding short-term interactions, Chang et al. measured salivary OT in cats before and after free interaction with their owners, finding significant OT increases in securely attached cats, with greater increases observed in individuals displaying more approach and hovering behaviors.[16] On the human side, Johnson et al. measured salivary OT in women before and after interacting with their pet cats, but found no significant overall change. However, specific behaviors–including petting the cat and cat-initiated approach–were positively correlated with OT change, suggesting that the quality and type of interaction modulates OT release.[17] Together, these findings suggest that similar affiliative behaviors, particularly cat-initiated approach and physical contact, may be associated with OT responses in both members of the dyad. To date, no study has simultaneously measured OT in both humans and cats, leaving the question of interspecies OT association unresolved.

From a long-term perspective, Nagasawa et al. reported that cats in households with more frequent owner-initiated tactile and vocal interaction had higher urinary OT concentrations, while no factors associated with urinary cortisol were identified.[18] Critically, this study measured only feline OT, and the extent to which cat ownership produces coordinated physiological changes in both humans and cats remains unknown. A further limitation of existing research is its reliance on questionnaires to assess daily interaction, which may not accurately reflect actual behavior. Objective measurement of physical proximity between owner and cat offers a more direct index of naturalistic interaction.

The present study aimed to examine the associations between hormonal indices—oxytocin and cortisol—in both humans and their pet cats, and to explore the possibility of interspecies physiological correlations. Specifically, we hypothesized that (1) owner and cat oxytocin and cortisol concentrations would be correlated within dyads, paralleling patterns reported in human-dog dyads; (2) hormonal concentrations in both owners and cats would be associated with owner mood, given the established roles of oxytocin and cortisol in affiliative and stress-related psychological states; and (3) closer daily physical proximity, objectively measured using BLE beacons, would be associated with more positive owner mood, whereas cohabitation duration alone would show weaker associations with owner mood. Finaly, as a supplementary analysis, we additionally employed the secure base test to exploratorily examine whether the hormonal indices identified above have functional behavioral correlates–specifically, whether cats' oxytocin and cortisol concentrations were associated with their approach behavior toward familiar (owner) and unfamiliar (stranger) individuals.

RESULTS

Descriptive statistics for all key variables are provided in Table S1. Full results of all Bayesian regression models are provided in Tables S2–S8.

Association between owner and cat hormone concentrations

The multivariate model converged without divergent transitions (all $\widehat{R} \leq 1.01$). The model included 24 cats from 17 owners with complete hormone data; for each individual, hormone concentrations were averaged across three urine samples collected over the one-month study period. Owner oxytocin concentration was positively associated with cat oxytocin concentration (β = 0.55, 95% CI [0.17, 0.91]; Figures 1A and 1B); owner cortisol concentration was negatively associated with cat oxytocin concentration (β = −0.47, 95% CI [−0.82, −0.12]; Figures 1A and 1C); both credible intervals excluded zero. In contrast, the associations of owner oxytocin with cat cortisol (β = −0.37, 95% CI [−0.80, 0.09]) and of owner cortisol with cat cortisol (β = 0.18, 95% CI [−0.24, 0.62]) did not exclude zero and are interpreted with caution given the exploratory nature of these analyses. The group-level standard deviations for owner identity were 0.52 (95% CI [0.03, 1.06]) for cat oxytocin and 0.47 (95% CI [0.02, 1.10]) for cat cortisol, indicating that clustering by owner accounted for a non-negligible proportion of variance comparable in magnitude to the residual variance (σ = 0.57 and 0.88, respectively). The correlation between the two owner-level random intercepts was estimated with wide uncertainty (r = 0.04, 95% CI [−0.92, 0.93]). The residual correlation between cat oxytocin and cat cortisol, after accounting for owner hormones and owner identity, was small and did not exclude zero (rescor = −0.04, 95% CI [−0.57, 0.51]). Full model output is provided in Table S2.

Associations between hormonal measures and owners' mood states

Owner cortisol concentration showed a consistent positive association with several negative mood dimensions (Figure 2), including Confusion-Bewilderment (posterior estimate = 0.75, 95% CI [0.49, 1.02]), Depression-Dejection (0.79, [0.53, 1.03]), Fatigue-Inertia (0.72, [0.43, 0.99]), Tension-Anxiety (0.74, [0.46, 1.01]), and Total Mood Disturbance (0.76, [0.50, 1.03]); all credible intervals excluded zero. The association with Vigor-Activity was also negative and credible (−0.43, [−0.79, −0.06]).

Cat oxytocin concentration was negatively associated with Anger-Hostility (−0.48, [−0.92, −0.04]), Confusion-Bewilderment (−0.55, [−0.90, −0.20]), Depression-Dejection (−0.39, [−0.75, −0.02]), Tension-Anxiety (−0.47, [−0.86, −0.08]), and Total Mood Disturbance (−0.43, [−0.81, −0.05]); all credible intervals excluded zero. Associations with Friendliness and Vigor-Activity included zero.

Owner oxytocin concentration was negatively associated with Anger-Hostility (−0.40, [−0.77, −0.03]); associations with the remaining six subscales and Total Mood Disturbance did not exclude zero. Cat cortisol concentration showed no credible associations with any POMS2 subscale.

As a sensitivity analysis, we additionally aggregated all variables to the household level (n = 19 owners for owner oxytocin/cortisol; n = 17 owners with valid cat hormone data for cat oxytocin/cortisol), since owner mood did not vary within household and models incorporating owner identity as a group-level effect on the dyad-level data were not identifiable (see STAR Methods). The associations of owner cortisol with negative mood, and of

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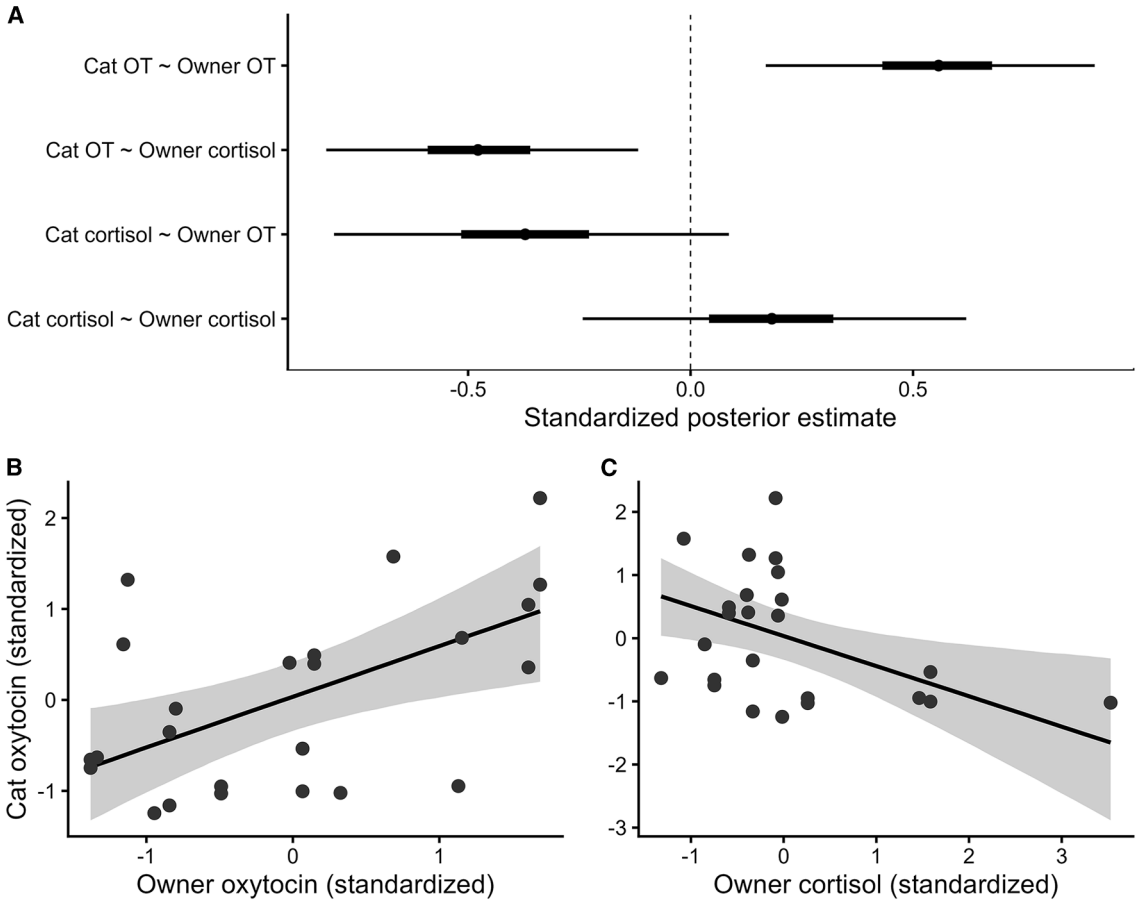


Figure 1. Hormonal associations between owners and their pet cats

A: Points show posterior medians; thick and thin lines show 50% and 95%credible intervals, respectively (stat_pointinterval, tidybayes). Dashed line indicates zero. Regression coefficients are from a single multivariate Bayesian model in which both owner hormones were entered as predictors of both cat hormones (see Table S2); paths are labeled by model formula and do not imply causal direction. B–C: Points show raw dyad-level data (n = 24). Lines and shaded bands show model-implied conditional effects and 95% credible intervals (conditional_effects, brms), holding the other owner hormone at its sample mean; they therefore correspond to the partial regression coefficients in A rather than simple bivariate relationships. B shows the association between owner and cat oxytocin concentrations, whereas C shows the association between owner cortisol and cat oxytocin concentrations.

cat oxytocin with Confusion-Bewilderment and Tension-Anxiety, remained credible in this sensitivity analysis; the remaining associations with cat oxytocin, and the association of owner oxytocin with Anger-Hostility, did not exclude zero after aggregation (Table S4).

Associations between owner mood states and owner–cat relationship variables

Closer owner-cat proximity was positively associated with Vigor-Activity (posterior estimate = 0.46, 95% CI [0.12, 0.82]; Figures 3A and 3B); associations with the remaining six POMS2 subscales and Total Mood Disturbance did not exclude zero. Longer cohabitation duration was positively associated with Confusion-Bewilderment (0.44, [0.09, 0.78]), Depression-Dejection (0.43, [0.07, 0.79]), Tension-Anxiety (0.43, [0.07, 0.79]), and Total Mood Disturbance (0.39, [0.03, 0.75]; Figures 3A and 3C); associations with Anger-Hostility, Fatigue-Inertia, Friendliness, and Vigor-Activity did not exclude zero.

In a household-level sensitivity analysis (n = 19 owners), the association between proximity and Vigor-Activity remained credible (0.55, [0.11, 0.99]), whereas none of the associations between cohabitation duration and owner mood were supported (all 95% credible intervals included zero; Table S6). These findings should be interpreted with caution given the exploratory, univariate nature of these analyses.

Associations between hormonal measures and circle entry duration in the SBT

Cat oxytocin concentration was associated with the probability of entering the circle area (hurdle component: posterior estimate = −1.88, 95% CI [−4.11, −0.20]), indicating that cats with higher oxytocin concentrations were more likely to approach and enter the circle. No credible associations were found for the duration spent inside the circle among cats that did approach, nor for the effects of target type or test phase.

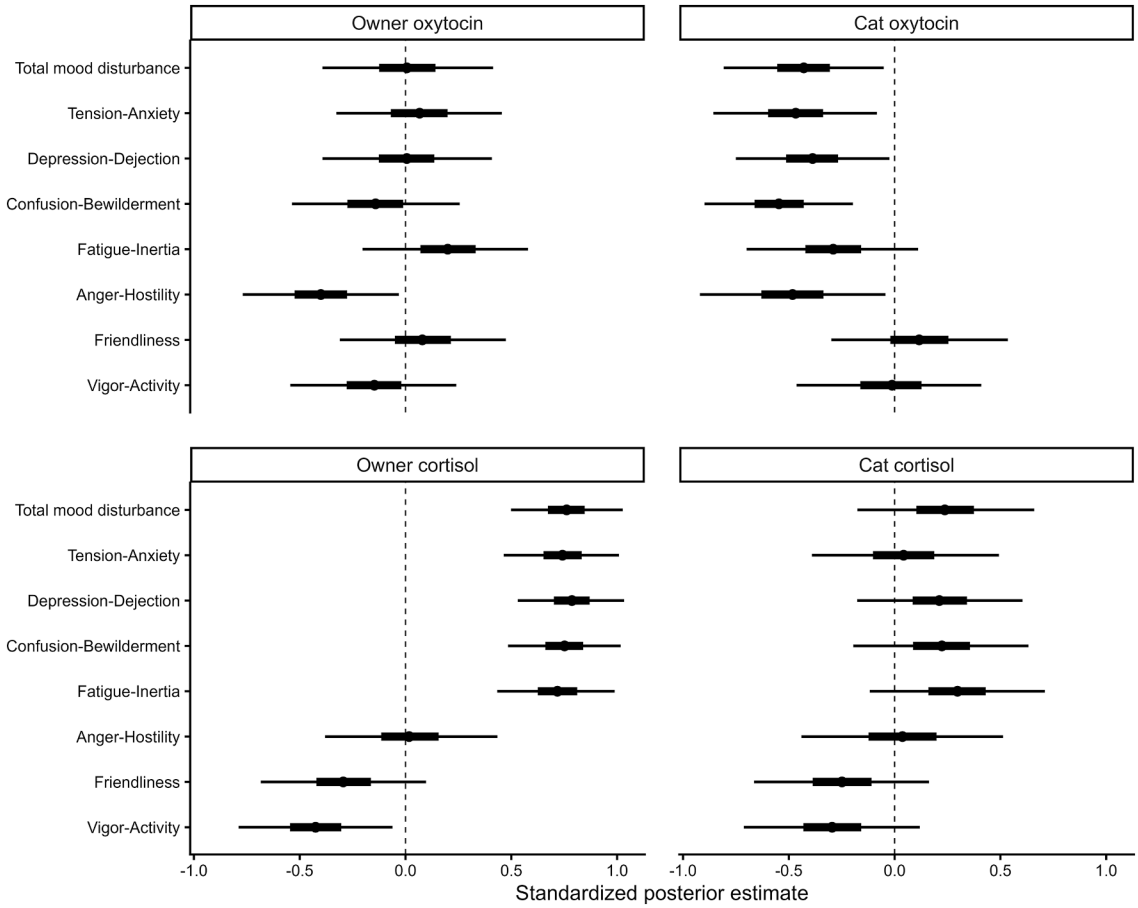


Figure 2. Associations between hormonal measures and owner mood states (POMS2 subscales), estimated at the level of individual owner-cat dyads

Posterior estimates from Bayesian regression models examining associations between hormonal measures and standardized POMS2 subscale scores. Separate models were fitted for owner oxytocin, cat oxytocin, owner cortisol, and cat cortisol concentrations (owner oxytocin and owner cortisol models: $n = 29$ dyads; cat oxytocin and cat cortisol models: $n = 24$ dyads with valid cat hormone data). Points indicate posterior median estimates, thick horizontal bars represent 50% credible intervals, and thin horizontal bars represent 95% credible intervals. Positive estimates indicate higher mood scores associated with higher hormone concentrations, whereas negative estimates indicate lower mood scores associated with higher hormone concentrations. Household-level sensitivity analyses are provided in Table S4.

Cat cortisol concentration showed a weak interaction with target type on the duration spent inside the circle (target [stranger vs. owner] × cat cortisol concentration: posterior estimate = −0.76, 95% CI [−1.54, 0.00]), such that cats with higher cortisol concentrations tended to spend less time near the stranger relative to the owner; however, this association was not clearly distinguishable from zero. No credible associations were found for the hurdle component (probability of approach) or for other main effects and interactions.

DISCUSSION

The present study examined associations between oxytocin and cortisol concentrations—each based on a single average value derived from three urine samples collected over a one-month period—in owners and their pet cats, with a particular focus on the possibility of interspecies physiological correlations. Using a multivariate Bayesian model, we jointly examined cat oxytocin and cortisol concentrations as correlated outcomes and estimated the associations of both owner oxytocin and owner cortisol with each feline hormone. The clearest associations involved cat oxytocin, whereas evidence for associations with cat cortisol was limited. Specifically, higher owner oxytocin concentrations were associated with higher cat oxytocin concentrations, whereas higher owner cortisol concentrations were associated with lower cat oxytocin concentrations. Cat oxytocin concentrations were, in turn, negatively associated with several dimensions of owner negative mood, whereas owner oxytocin and cat cortisol concentrations showed comparatively weak associations with owner mood. In addition, among several exploratory associations examined between daily proximity, cohabitation duration, and owner mood, only proximity showed a credible positive association with Vigor–Activity scores. Exploratory behavioral analyses further suggested that higher cat oxytocin

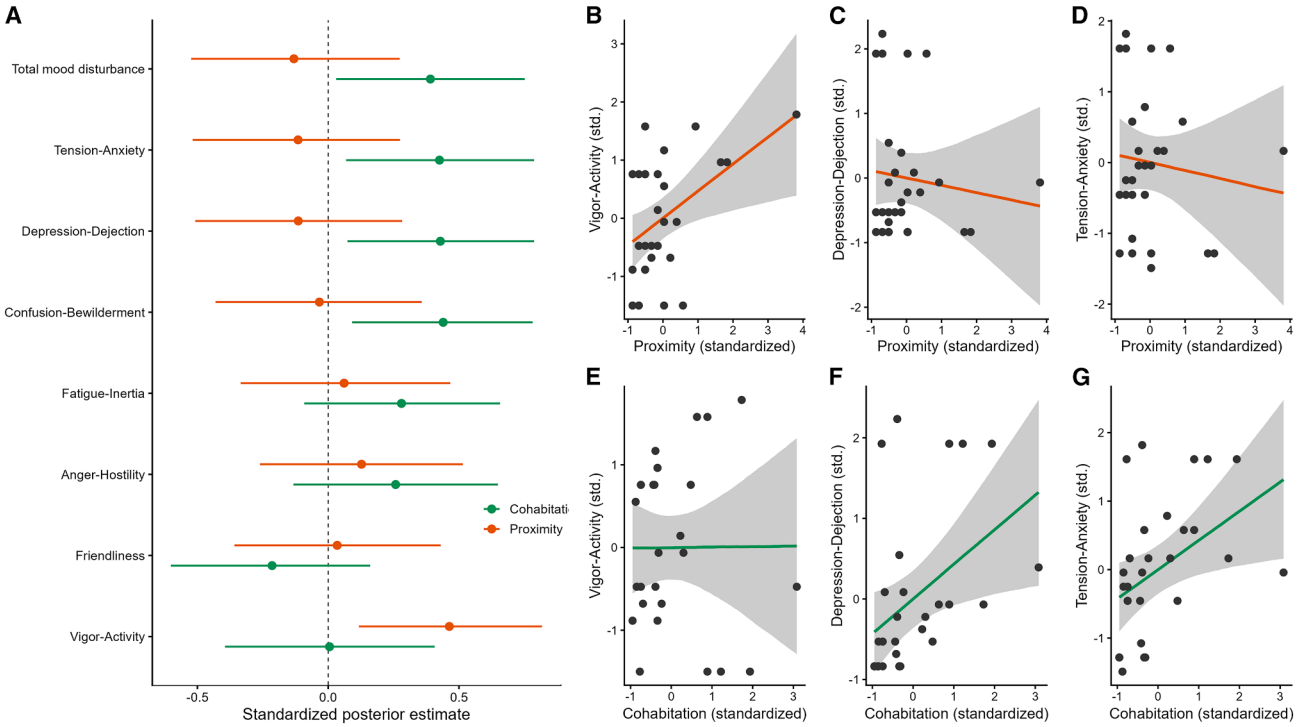


Figure 3. Associations of proximity and cohabitation duration with owner mood (POMS2 subscale scores), estimated at the level of individual owner-cat dyads (*n* = 29)

(A) Standardized posterior estimates for cohabitation duration (green) and proximity (orange) across the seven POMS2 subscales and Total Mood Disturbance. Points show posterior medians; horizontal lines show 95% credible intervals. Dashed line indicates zero. Regression coefficients are from separate univariate Bayesian models for each predictor and each subscale (see STAR Methods); proximity was measured as the monthly median RSSI during periods when owners were at home.

(B) Model-implied conditional effects of proximity on Vigor-Activity, Depression-Dejection, and Tension-Anxiety.

(C) Model-implied conditional effects of cohabitation duration on the same three subscales. Points show dyad-level data; lines and shaded bands show posterior means and 95% credible intervals from the corresponding univariate models (conditional_effects, brms). Among these six associations, proximity with Vigor-Activity and cohabitation duration with Depression-Dejection and Tension-Anxiety had 95% credible intervals excluding zero (A); the remaining three are shown for direct comparison across predictors and subscales. Associations involving cohabitation duration were not supported in household-level sensitivity analyses (Table S6). Abbreviations: TMD, Total Mood Disturbance; TA, Tension-Anxiety; DD, Depression-Dejection; CB, Confusion-Bewilderment; FI, Fatigue-Inertia; AH, Anger-Hostility; F, Friendliness; VA, Vigor-Activity.

concentrations were associated with increased approach-related behavior, whereas higher cortisol concentrations were linked to reduced behavior toward strangers. Together, these findings suggest that human–cat relationships may involve cross-sectional hormonal associations centered primarily on the oxytocin system and associated with everyday social interaction.

The present study is, to our knowledge, the first to demonstrate a positive cross-sectional association between owner and cat oxytocin concentrations in a naturalistic home setting. Oxytocin is a neuropeptide classically associated with mother-infant bonding and social attachment,[19] and the present findings raise the possibility that cats, when forming affiliative relationships with humans, may recruit oxytocin-mediated bonding mechanisms analogous to those underlying maternal attachment. Oxytocin synchrony in the human-dog dyad has been demonstrated primarily in the context of specific affiliative behaviors: mutual gazing has been shown to activate an oxytocin positive-feedback loop between owners and dogs,[13] and the frequency of owner-initiated kissing has been positively correlated with dog oxytocin levels.[20] The present findings suggest that a similar affiliative oxytocin linkage exists in the human-cat dyad. Oxytocin release is broadly associated with physical contact and affiliative touch,[9] and cat-initiated approach and physical contact with owners have been shown to correlate with oxytocin changes in both cats and owners,[16,17] suggesting that tactile interaction may be a primary driver of oxytocin release in human-cat dyads. Beyond physical contact, however, it is also possible that non-contact affiliative signals play a role: cats have been shown to engage in affiliative communication with humans through slow eye blinking–a behavior interpreted as a signal of relaxed, positive affect[21]—raising the possibility that, analogous to the gaze-mediated oxytocin loop in human-dog dyads, oxytocin release in both members of human-cat dyads may also be partially facilitated through visual affiliative signals without direct physical contact. Notably, oxytocin in cats may function differently depending on the social context. Koyasu et al. reported that in group-housed cats, oxytocin was

negatively associated with inter-cat contact–the opposite of its typical affiliative role in group-living species.[22] Combined with the present findings, this raises the speculative possibility that domestication may have reshaped the oxytocinergic system in cats to favor human-directed over conspecific affiliation; however, given the correlational nature of both studies, this interpretation remains untested and would require comparative or experimental work directly contrasting human-directed and conspecific oxytocin responses within the same cats.

The multivariate approach also enabled us to examine hormone-specific and cross-hormonal associations without combining oxytocin and cortisol into a single composite index. Higher owner oxytocin and lower owner cortisol concentrations were both associated with higher cat oxytocin concentrations, suggesting that cat oxytocin may be linked to a dyadic physiological context characterized by affiliative and relatively low-stress states in owners. One possible explanation is that frequent affiliative tactile interactions, such as petting, may promote oxytocin release in cats while attenuating cortisol levels in owners. However, because the present study was cross-sectional and did not measure hormonal changes before and after specific interactions, this interpretation remains speculative.

Notably, the positive association between owner and cat oxytocin concentrations, together with the limited evidence for an association between owner and cat cortisol concentrations, contrasts with findings from dog research, where cortisol associations have been more consistently documented, both in short-term contexts such as agility competition[23] and search-and-rescue tasks,[24] and over longer time periods as indexed by hair cortisol concentrations.[15] Cortisol synchrony in dog-owner dyads is thought to be mediated by emotional contagion–the mirroring of stress states between individuals–supported by behavioral and physiological evidence including heart rate variability synchronization during shared tasks,[25] and by findings that dogs can discriminate human stress odors and that such odors influence their emotional states.[26,27] The limited evidence for an owner–cat cortisol association in the present study may reflect fundamental differences in the human-cat relationship relative to the human-dog relationship. Höglin et al. suggested that cortisol synchrony in dogs may be characteristic of breeds selected for close cooperation with humans and is strongly modulated by relationship quality[28]–conditions that differ markedly in cats, which have undergone a distinct domestication history and maintain greater behavioral independence. This contrast may also reflect differing social ecologies: the domestic dog's ancestor is a highly group-living species, whereas the domestic cat has retained a largely solitary or facultatively social organization,[1] such that cortisol regulation tied to ongoing group-level social dynamics may be less relevant to a species that evolved largely outside such contexts. Although cats can discriminate human emotional odors and show behavioral responses to them,[29] this sensitivity does not appear to translate into cortisol correlation, a pattern consistent with both the absence of meaningful associations between cat cortisol and owner mood in the present study, and with a previous finding of no significant correlation between hair cortisol concentrations in cats and their owners.[30] It should also be noted that the cited dog studies relied on markedly different sampling approaches—including hair cortisol reflecting cumulative concentrations over several months, and saliva or serum cortisol measured around specific short-term events—whereas the present study assessed urinary cortisol averaged across three samples collected over one month; this methodological divergence limits the extent to which species differences can be cleanly inferred from this comparison.

A notable finding was that cat oxytocin concentrations were negatively associated with multiple dimensions of owner negative mood, including Anger–Hostility, Confusion–Bewilderment, Depression–Dejection, Tension–Anxiety, and Total Mood Disturbance. This suggests that cats with higher oxytocin levels–which may reflect a more affiliative and socially engaged disposition–tend to live with owners who report better psychological well-being. Notably, cat oxytocin concentrations showed associations with a broader range of mood dimensions than did owner oxytocin concentrations, which showed a credible association only with Anger-Hostility. This asymmetry may reflect the fact that human oxytocin levels are influenced by a wide range of social and biological factors beyond the owner-cat relationship–including interactions with other people, physical contact, and reproductive status–whereas cat oxytocin may more exclusively reflect the quality and frequency of interactions with their owner. In this sense, cat oxytocin may serve as a more sensitive index of the relational quality within the dyad than owner oxytocin; however, this interpretation remains speculative, and alternative explanations cannot be excluded. For instance, the stronger association observed for cat oxytocin could also reflect lower measurement variability in feline oxytocin concentrations, or fewer uncontrolled sources of influence on the feline oxytocinergic system relative to the human system, rather than a genuine difference in relational sensitivity. Furthermore, it is possible that cat oxytocin influences owner mood indirectly through behavior: cats with higher oxytocin concentrations tend to initiate more approach and affiliative behaviors,[16] which may in turn promote positive mood states in the owner.

Conversely, higher owner cortisol concentrations were positively associated with negative mood dimensions, consistent with the well-established role of cortisol as a marker of psychological stress.[7,12] Taken together, these findings suggest that the hormonal state of the cat may be meaningfully linked to the mental health of the owner, and that the oxytocin system in particular may play a role in linking the emotional states of cats and their owners. However, given the cross-sectional and correlational nature of the present study, the directionality of these associations remains unclear. It is possible that owners in better psychological health engage in more affiliative interactions with their cats, thereby promoting oxytocin release in the cat; alternatively, living with a cat displaying higher oxytocin levels–and thus more approach and affiliative behavior–may itself contribute to improved owner psychological health. Indeed, cat-initiated approach behavior has been associated with oxytocin increases in cats[16] and positively correlated with oxytocin changes in owners,[17] suggesting that the cat's social initiative may be a key driver of affiliative hormonal responses in both members of the dyad.

The present study also examined associations between daily proximity, cohabitation duration, and owner mood states.

Table 1. Characteristics of the participating owners (n = 19)

Owner ID	Age (years)	Sex
H01	31	female
H02	38	female
H03	58	female
H04	36	female
H05	23	female
H06	30	female
H07	28	female
H08	51	female
H09	48	female
H10	60	male
H11	28	male
H12	18	female
H13	51	female
H14	60	female
H15	47	female
H16	49	female
H17	25	female
H18	38	female
H19	27	male

Values are shown per owner. Mean age ± SD: 39.3 ± 13.4 years (range 18–60); Female: 16 (84.2%), Male: 3 (15.8%).

Among the 16 associations examined, closer owner-cat proximity, as objectively measured by BLE beacons, was positively associated with Vigor-Activity scores, and longer cohabitation duration was positively associated with Confusion-Bewilderment, Depression-Dejection, Tension-Anxiety, and Total Mood Disturbance; associations with the remaining subscales did not exclude zero. Given the exploratory and univariate nature of these analyses, these associations should be interpreted with caution. Notably, the association between cohabitation duration and owner mood was not supported in a household-level sensitivity analysis (see Results), suggesting that it may partly reflect the non-independence of cats from the same household rather than a robust relationship-level effect; the association between proximity and Vigor-Activity, by contrast, remained credible in both analyses. If the proximity association is robust, it would be consistent with previous research demonstrating short-term psychological effects of human-cat interaction, such as increased emotional arousal following interaction with cats,[31] and would suggest that objective, continuous measurement of proximity may complement the limitations of self-report questionnaires in capturing naturalistic daily interaction.

Of note, proximity and cohabitation duration were associated with qualitatively different aspects of owner mood: proximity was specifically associated with higher Vigor-Activity (a positive-affect dimension), whereas cohabitation duration was associated with several negative mood dimensions. This divergent pattern between cohabitation duration and proximity may help reconcile the inconsistent effects of cat ownership reported in the literature: cat ownership has shown weaker, and at times negative, associations with owner well-being relative to dog ownership,[4,6] and benefits that do emerge appear to fade rapidly after acquisition.[5] Cohabitation duration and cat-ownership status, as used in most prior studies, capture only whether and how long an owner has lived with a cat, not the nature or quality of their daily interaction with it. The present finding suggests that time spent in close physical proximity to one's cat—plausibly reflecting the frequency of petting, play, or other direct engagement—may be associated with more immediate, positive affective states, whereas cohabitation duration alone may be confounded with other factors that are associated with negative mood. One candidate confound is cat age, which was strongly correlated with cohabitation duration in this sample (r = 0.89). However, leave-one-out cross-validation comparing models with cohabitation duration alone, cat age alone, and both predictors together showed that the cohabitation-duration-only model best predicted all four affected subscales, indicating that the association is not attributable to cat age (Table S8). If so, inconsistencies across previous studies of cat ownership and well-being may partly reflect unmeasured variation in the quality and quantity of daily owner-cat interaction, rather than a genuine absence of psychological benefit from cat ownership. The exploratory analyses using the Secure Base Test provided complementary behavioral evidence consistent with the hormonal findings. Cats with higher oxytocin concentrations were more likely to enter the circle area during the SBT, regardless of whether the target individual was the owner or a stranger. This finding aligns with the interpretation that oxytocin reflects a more generally affiliative and socially engaged disposition in cats, rather than owner-specific attachment per se. This is consistent with Chang et al., who reported that cats with higher oxytocin concentrations following owner interaction showed more approach and hovering behaviors,[16] suggesting that oxytocin may index a broader social motivation in cats that facilitates proximity-seeking across social partners.

In contrast, cats with higher cortisol concentrations tended to spend less time in the stranger-associated area, suggesting that chronic stress or anxiety may selectively inhibit approach behavior toward unfamiliar individuals while leaving owner-directed behavior relatively intact. This pattern mirrors findings across vertebrate species showing that elevated glucocorticoids are associated with reduced social approach toward conspecifics,[32] as well as evidence in humans demonstrating that higher cortisol stress responses are associated with reduced active approach toward social stimuli.[33] Taken together, these behavioral findings suggest that oxytocin and cortisol may play functionally distinct roles in shaping cat social behavior: oxytocin facilitating general affiliative motivation, and cortisol modulating the threshold for approach toward unfamiliar individuals.

Limitations of the study

Several limitations of the present study should be noted. First, this study was exploratory and not pre-registered; therefore, interpret the analyses with caution, and replicate the findings in future confirmatory studies. Second, the sample size was relatively small, with 29 owner-cat dyads drawn from 19 owners, which limits the statistical power and generalizability

Table 2. Characteristics of the participating cats (*n* = 29)

Cat ID	Age (months)	Cohabitation duration (months)	Sex	Breed	Neutered/spayed	Urine sample
C01	25	23	male	mixed	yes	yes
C02	77	57	male	Ragamuffin	yes	yes
C03	25	21	female	Persian	yes	yes
C04	25	25	male	mixed	yes	yes
C05	53	22	female	Exotic Shorthair	yes	yes
C06	120	114	male	American Shorthair	yes	no
C07	90	73	male	mixed	yes	yes
C08	88	86	female	American Shorthair	yes	no
C09	10	8	female	mixed	no	yes
C10	162	106	female	mixed	yes	yes
C11	28	29	male	Maine Coon	yes	yes
C12	13	11	female	mixed	yes	yes
C13	84	47	male	unknown	yes	yes
C14	16	9	male	mixed	yes	yes
C15	16	9	male	mixed	yes	yes
C16	9	4	female	mixed	yes	yes
C17	119	23	female	Scottish Fold	yes	yes
C18	11	9	male	mixed	yes	no
C19	7	5	male	mixed	yes	no
C20	29	26	male	Ragdoll	yes	yes
C21	8	5	female	mixed	yes	yes
C22	27	25	male	Chartreux	yes	yes
C23	27	25	male	Chartreux	yes	yes
C24	87	50	male	mixed	yes	yes
C25	66	63	male	Norwegian Forest Cat	yes	yes
C26	74	73	male	mixed	yes	yes
C27	161	159	male	mixed	yes	no
C28	4	1	male	mixed	yes	yes
C29	4	1	male	mixed	yes	yes

Values are shown per cat. Age: mean ± SD = 50.5 ± 47.0 months (range 4–162); cohabitation: mean ± SD = 38.2 ± 39.2 months; male: 20 (69.0%), female: 9 (31.0%); neutered/spayed: 28 (96.6%).

of the findings. Relatedly, because several owners contributed more than one cat, observations were not fully independent. Models including owner identity as a group-level effect were not identifiable, and although household-level sensitivity analyses are provided (Tables S4 and S6), some residual non-independence remains. Third, measuring daily proximity using BLE beacons has several limitations. RSSI-based distance estimation may have been affected by variations in home layout and construction materials, as it is known to be susceptible to environmental interference.[34] In addition, because the receiving device (a smartphone) was active only while owners were at home and its on/off timestamps were not logged, the total duration of home time could not be verified; proximity is therefore reported as the monthly median RSSI during detected contact, rather than as the total duration or proportion of time spent near the owner. Fourth, owners and cats shared the same household environment throughout the study. Therefore, the observed associations between their hormone concentrations may partly reflect shared environmental factors, such as household noise, crowding, temperature, daily routines, or other contextual conditions, rather than dyad-specific social interactions. Relatedly, the human–cat interactions that occurred within the circle during the secure base test (e.g., verbal greeting, petting, eye contact) were not standardized and were left to occur naturally ("interact as you normally would"), which may have introduced variability between owners and strangers and could have influenced the time cats spent near each person. The present analyses therefore rely solely on time spent within the circle and do not capture the quality or nature of these interactions; future studies could employ a structured behavioral ethogram to quantify and standardize dyadic interactions during the test. Fifth, owners were recruited voluntarily through social media and flyers, and the sample was predominantly female (16 of 19 owners). Given potential sex-related differences in human OT concentrations[35] and OT-related responsesM,[36] this imbalance may limit the generalizability of the hormonal findings to male owners. We did not record household composition or the presence of additional caregivers, which may have

influenced the cats' social and hormonal environment. These factors limit the generalizability of the findings and should be addressed in future studies using more diverse samples and detailed household information. Sixth, the timing and temporal resolution of the psychological, hormonal, and proximity measurements were not aligned. POMS2 was administered once, hormone concentrations were averaged across three urine samples collected over one month, and proximity was summarized from continuous recordings across the one-month study period. In addition, urine sampling schedules differed between species: owner samples were collected as first morning voids, whereas cat samples were collected opportunistically without restriction on the time of day. This difference may have introduced additional variability, particularly in cat cortisol concentrations, and may have reduced our ability to detect owner–cat cortisol associations. Although averaging three samples may have reduced some short-term variability, it could not fully account for the different sampling schedules. Accordingly, these findings should be interpreted as exploratory cross-sectional associations among aggregate measures rather than evidence of concurrent changes. Future studies should use repeated, concurrent, and temporally standardized measurements from both members of the dyad. Finally, given the cross-sectional and correlational design of the present study, the directionality of the observed associations cannot be determined. Longitudinal or experimental designs will be necessary to clarify the causal mechanisms underlying the hormonal associations observed between owners and their cats.

Despite these limitations—including the possibility that the observed hormonal correlations partly reflect shared household environments rather than dyad-specific interactions—, the present findings provide preliminary evidence that owner and cat hormonal states are associated primarily through oxytocin-related processes, whereas evidence for an association between their cortisol concentrations was limited. Although the mood-enhancing effects of cat ownership may diminish over time, daily interaction and physical proximity with cats may help sustain positive psychological states in owners–suggesting that cats serve not merely as passive companions but as active social partners capable of participating in bidirectional physiological and emotional interactions with humans.

RESOURCE AVAILABILITY

Lead contact

Further information and requests for resources should be directed to and will be fulfilled by the lead contact, Saho Takagi (takagi.saho.5x@kyoto-u.ac.jp).

Materials availability

This study did not generate new unique materials.

Data and code availability

- Data: The raw data used for all analyses reported in this paper are provided as Data S1 and are publicly available at https://github.com/SahoTakagi/cat-owner-hormone-2026 (listed in the key resources table).
- Code: All R code used for the statistical analyses reported in this paper is publicly available at https://github.com/SahoTakagi/cat-owner-hormone-2026 (listed in the key resources table).
- Any additional information required to reanalyze the data reported in this paper is available from the lead contact upon request.

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AUTHOR CONTRIBUTIONS

S.T. and T.T. collected data. S.T. performed statistical analyses and wrote the manuscript. M.N. and T.K. conceptualized the study together with S.T. and T.T., supervised the overall study design, contributed to data interpretation, and critically reviewed the manuscript. The dataset used in this study is based on the Master's thesis of T.T. All authors read and approved the final version of the manuscript.

DECLARATION OF INTERESTS

The authors declare no competing interests.

DECLARATION OF GENERATIVE AI AND AI-ASSISTED TECHNOLOGIES IN THE MANUSCRIPT PREPARATION PROCESS

During the preparation of this work, the author(s) used Claude (Anthropic) to assist with manuscript drafting, translation, and editing. The author(s) reviewed and edited all output as needed and take full responsibility for the content of the published article.

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SUPPLEMENTAL INFORMATION

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STAR★METHODS

KEY RESOURCES TABLE

REAGENT or RESOURCE	SOURCE	IDENTIFIER
Antibodies		
Mouse IgG-Fc Fragment Antibody	Bethyl Laboratories	Cat# A90-131A; RRID: AB_67172
Anti-cortisol primary antibody	Abcam	Cat# ab1949; RRID: AB_302703
Biological samples		
Cat urine samples	This paper	N/A
Human (owner) urine samples	This paper	N/A
Critical commercial assays		
Cortisol EIA (in-house, adapted protocol)	This paper; adapted from ref[37]	N/A
Oxytocin competitive ELISA (AirPlantsBio-Original)	AirPlantsBio Co., Ltd.	N/A
Chemicals, peptides, and recombinant proteins		
Urea hydrogen peroxide (Substrate buffer A)	Sigma-Aldrich	Catalog# V1753
Citric acid monohydrate (Substrate buffer A/B)	FUJIFILM Wako	Catalog# 035-03495
TMBZ (Substrate buffer B)	FUJIFILM Wako	Catalog# 346-04031
DMSO (Substrate buffer B)	Sigma-Aldrich	Catalog# D5879
Tween 80 (Washing buffer)	Sigma-Aldrich	Catalog# P1754
BSA, Fraction (Assay buffer)	Sigma-Aldrich	Catalog# A7030
Sulfuric acid (4N, diluted from 36N)	Kanto Chemical Co., Ltd.	Catalog#37390-00
Cortisol-HRP conjugate	MyBioSource	Catalog# MBS342079
Synthetic oxytocin (s-OT standard)	Peptide Institute, Inc.	Catalog# 4084V
Picric acid (TNP solution)	FUJIFILM Wako Pure Chemical Corporation	Catalog#205-08672
Sodium hydroxide (NaOH)	Kanto Chemical Co., Inc.	Catalog#37184-00
Creatinine standard	Kanto Chemical Co., Inc.	Catalog#033-04591
Deposited data		
Raw data (Data S1)	This paper	Data S1; github.com/SahoTakagi/cat-owner-hormone-2026
R code for all statistical analyses	This paper	github.com/SahoTakagi/cat-owner-hormone-2026
Experimental models: Organisms/strains		
Domestic cat (Felis catus), $n = 29$	This paper	N/A
Software and algorithms		
R (version 4.4.1)	R Core Team	r-project.org
brms (version 2.21.0)	Bürkner, 2017	paul-buerkner.github.io/brms
Other		
96-well high-binding microplates	Costar	Catalog# 9018
MaxiSorp immunoplates	Nunc	Catalog# 469914
BLE beacon	Access	FBV1004KH3-ACS
Custom RSSI-logging smartphone application	This paper	N/A (available from lead contact upon request)
POMS2 short form (Japanese version)	Heuchert et al.[38]	N/A

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EXPERIMENTAL MODEL AND STUDY PARTICIPANT DETAILS

This study was approved by the Institutional Animal Care and Use Committee of Azabu University (approval number: #2499069) and the Human Research Ethics Committee of Azabu University (approval number: #097). All procedures involving animals were conducted in accordance with the ethical standards of the institution. Written informed consent was obtained from all human participants prior to their participation in the study.

Owners

Owners were recruited on an ongoing basis through social media and printed flyers. Individuals who volunteered and provided informed consent were enrolled in the study, and no formal exclusion criteria were applied. Nineteen cat owners participated in this study (16 females, 3 males; age range: 18–60 years; mean age ± SD = 39.3 ± 13.4 years). All participants were the primary caregivers of their cats and resided with them in a domestic environment. The number of cats per owner ranged from one to four: eleven owners had one cat, seven owners had two cats, and one owner had four cats (Table 1).

Cats

Twenty-nine domestic cats participated in this study (20 males, 9 females; age range: 4–162 months; mean age ± SD = 50.5 ± 47.0 months at the start of the experiment; mean duration of ownership ± SD = 38.2 ± 39.2 months). Breeds included mixed-breed cats (n = 17), American Shorthair (n = 2), Chartreux (n = 2), Ragamuffin (n = 1), Persian (n = 1), Exotic Shorthair (n = 1), Maine Coon (n = 1), Scottish Fold (n = 1), Ragdoll (n = 1), Norwegian Forest Cat (n = 1), and one cat of unknown breed. The majority of cats were neutered or spayed (n = 28; 96.6%), with one sexually intact female (C09). All cats were housed indoors with their owners. In total, 29 owner-cat dyads were included in the analyses (Table 2). We did not assess cats' prior exposure to unfamiliar people or their general sociability, which may have influenced both hormone concentrations and approach behavior toward the stranger.

METHOD DETAILS

Study procedure overview

The study was conducted over a one-month period. During this time, each owner-cat dyad completed the following assessments: (1) collection and submission of urine samples from both the owner and the cat for hormonal analysis of oxytocin and cortisol; (2) continuous monitoring of owner-cat physical proximity using Bluetooth Low Energy (BLE) beacons; (3) the Japanese short form of the Profile of Mood States 2 (POMS2) to assess owner mood state; and (4) a Secure Base Test (SBT) to exploratorily examine the functional behavioral correlates of hormonal indices. Details of each procedure are described below.

Urine sample collection

Urine samples were collected from both owners and their cats on three occasions during the one-month study period. Three collections were selected as a feasible repeated-sampling schedule because the timing of spontaneous urination in cats is difficult to predict and home collection is challenging. Human first-morning urine was collected using a collection cup and transferred to a tube using a dropper. Cat urine was collected during spontaneous urination by holding a cotton swab directly in the urine stream. Owner and cat urine samples were not necessarily collected on the same dates. Because this method required an observer to be present at the moment of spontaneous urination, a sample could not always be obtained within the collection window; consequently, urine samples were not collected for 5 of the 29 cats. Missingness did not differ systematically between single- and multi-cat households, nor by breed or sex, and is therefore considered unrelated to specific cat or household characteristics.

Each sample was stored in the owner's home freezer until all three samples had been collected. The samples were then shipped together to the laboratory by frozen courier at approximately −15°C and stored at −80°C upon arrival. Before analysis, samples were thawed on ice, and the urine absorbed in the cotton swabs was recovered by centrifugation at 3,500 rpm at 4°C for 15 min. Human urine samples underwent the same storage, shipping, thawing, and centrifugation procedures after collection.

Hormonal analysis

Urine samples collected from both owners and cats were analyzed for creatinine, cortisol, and oxytocin concentrations using enzyme-linked immunosorbent assay (ELISA). Creatinine concentrations were measured to correct for urine dilution, and all hormonal concentrations were expressed as ratios relative to creatinine (ng/mg creatinine).

Creatinine. Urinary creatinine concentrations were determined using a colorimetric assay based on the Jaffe reaction. Urine samples (undiluted for humans and 4-fold diluted for cats) and creatinine standards were dispensed at 100 μL/well into 96-well microplates. Subsequently, 50 μL each of 1 M NaOH and picric acid reagent (TNP) were added to each well. After incubation for 20 min at room temperature, absorbance was measured at 490 nm.

Cortisol. Cortisol concentrations were measured using an in-house enzyme immunoassay (EIA) adapted from a previously described protocol.[37] Briefly, 96-well high-binding microplates (Costar, Cat# 9018) were coated with Mouse IgG-Fc Fragment Antibody (Bethyl, A90-131A) and blocked with assay buffer containing 0.1% BSA. Undiluted urine samples and cortisol standards (serial 2-fold dilutions; 50–0.049 ng/mL) were incubated overnight at 4°C with cortisol-HRP conjugate (MBS342079; 1:100,000)

and anti-cortisol primary antibody (Abcam, ab1949; 1:200,000). Following TMB substrate development, absorbance was measured at 450 nm. All samples to be compared were run on the same plate.

Oxytocin. Urinary oxytocin concentrations were measured using an in-house competitive ELISA (AirPlantsBio-Original), with analysis conducted by AirPlantsBio Co., Ltd. Urine samples were concentrated or diluted prior to the assay: human urine was concentrated 4-fold by acetone precipitation, while cat urine was diluted 20-fold in PBS. Synthetic oxytocin (2 ng/mL) was coated onto MaxiSorp immunoplates (Nunc, 469914), and samples were premixed with a primary anti-oxytocin antibody (final dilution 1:15,500) before application in duplicate. A standard curve was constructed using serial 2-fold dilutions (2,000–7.8 pg/mL). Following incubation with an HRP-conjugated secondary antibody and TMB substrate development, absorbance was measured at 450 nm.

Measurement of daily proximity

To assess inter-individual distance between cats and their owners within the home environment, cats were equipped with small Bluetooth Low Energy (BLE) beacons (FBV1004KH3-ACS, Access) continuously for one month, 24 hours per day. BLE beacons estimate the distance between the beacon and a receiver unit based on Bluetooth signal strength (RSSI). A dedicated smartphone application was developed in-house to continuously record RSSI values between the cat's beacon and the owner's smartphone, which served as the receiver unit. Beacons were attached to cats via their collars in one of two ways: either by placing the beacon inside a purpose-made protective case mounted on a standard collar, or by fitting the beacon into a modified bandana-style collar. Each owner was provided with a smartphone preinstalled with the dedicated application, which served as the receiver unit for the beacon signals. Owners were instructed to carry the smartphone with them while at home. When leaving the house, they were asked to turn off the smartphone and leave it near the entrance, and to turn it back on upon returning home. To enable measurement of owner–cat proximity during the night, owners were also instructed to charge the smartphone near their usual sleeping position each night.

RSSI values were recorded continuously throughout the experimental period as an index of the physical distance between the owner's smartphone and the cat's beacon. As RSSI values below −90 dBm do not reliably correlate with distance, only values of −90 dBm or above were retained for analysis; values below this threshold were excluded as they were considered to reflect the maximum detectable range or signal loss. Because Bluetooth RSSI is subject to substantial measurement noise, the median (rather than mean) was used as a more robust summary statistic. The median RSSI value across the experimental period was used as an index of daily proximity between each owner and cat, with higher values indicating closer physical proximity.

Profile of Mood States 2 (POMS2)

To assess owners' mood states, we administered the Japanese short form of the Profile of Mood States 2 (POMS2).[38] Owners completed the questionnaire once at a self-selected time point within a one-month period, rating how they had felt over the past two weeks including the day of assessment. The short form consists of 35 items rated on a 5-point Likert scale, yielding scores for seven subscales: Anger-Hostility (AH), Confusion-Bewilderment (CB), Depression-Dejection (DD), Fatigue-Inertia (FI), Tension-Anxiety (TA), Friendliness (F), and Vigor-Activity (VA), plus an overall Total Mood Disturbance (TMD) score. The timing of the POMS2 assessment was not standardized relative to the urine collection dates.

Secure Base Test (SBT)

To exploratorily examine whether the hormonal indices identified in this study have functional behavioral correlates, we administered a Secure Base Test (SBT) adapted from Vitale et al.[39] investigating whether cats' oxytocin and cortisol concentrations were associated with their approach behavior toward the owner and a stranger. Unlike the original protocol in which the owner serves as the sole social partner, we additionally included a stranger as a target individual to enable direct comparison between familiar and unfamiliar target individuals. A single male experimenter (T.T., the second author) served as the stranger for all owner–cat dyads. The order of conditions (owner-first vs. stranger-first) was counterbalanced across dyads.

Each SBT consisted of two conditions (owner and stranger), each comprising three phases of two minutes each, for a total test duration of six minutes per condition. A circle of 1 m in diameter was marked on the floor to define the interaction zone. In the Baseline phase, the target individual sat at the center of the circle. When the cat entered the circle, the target individual engaged in interaction with the cat. In the Alone phase, the target individual said goodbye to the cat and left the room, leaving the cat alone. In the Return phase, the target individual re-entered the room and resumed the same procedure as in the Baseline phase. Throughout the Baseline and Return phases, target individuals were instructed to ignore the cat completely–including avoiding eye contact–when the cat was outside the circle, and to initiate interaction (e.g., verbal greeting, petting, eye contact) only when the cat entered the circle.

QUANTIFICATION AND STATISTICAL ANALYSIS

Inter-rater reliability

To assess inter-rater reliability, a random subset of videos (approximately 20% of total coded observations) was independently coded by a second rater blind to the study hypotheses. Intraclass correlation coefficients (ICC) were calculated using a two-way agreement model. Results indicated excellent inter-rater reliability (ICC = 0.991, 95% CI [0.969, 0.997]).

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Handling of missing data and sample sizes

Urine samples could not be obtained from five cats due to practical difficulties in sample collection. Consequently, hormonal analyses involving cat oxytocin and cortisol concentrations were conducted on the remaining 24 owner-cat dyads. All other analyses (beacon proximity, POMS2, and SBT) were conducted using the full sample of 29 owner-cat dyads unless otherwise noted. For each individual, hormone concentrations were averaged across the available urine samples to obtain a single hormone concentration index representing the one-month study period.

Standardization procedure

Continuous hormonal variables (oxytocin and cortisol concentrations) were standardized (z-scored) separately for each variable and each analysis, using the mean and standard deviation calculated from the subset of dyads or households with valid (non-missing) values for that variable in the corresponding analysis. Standardization was performed independently for owner and cat measures—that is, owner oxytocin and cat oxytocin were each centered and scaled using their own respective sample mean and SD, rather than relative to a shared or pooled distribution across owners and cats. The same procedure was applied to cortisol concentrations for the analyses involving $n = 24$ dyads, and to beacon proximity, cohabitation duration, and POMS2 subscale scores for the analyses involving the full sample ($n = 29$ dyads).

Owner–cat hormonal associations

To examine owner–cat hormonal associations without constructing a composite balance index, we fitted a multivariate Bayesian Gaussian regression model in which standardized cat oxytocin and cat cortisol concentrations were modeled jointly as correlated response variables. Standardized owner oxytocin and owner cortisol concentrations were entered simultaneously as predictors of both response variables. The model therefore estimated four regression associations: owner oxytocin with cat oxytocin, owner cortisol with cat oxytocin, owner oxytocin with cat cortisol, and owner cortisol with cat cortisol. Residual correlation between the two feline hormone outcomes was estimated using set_rescor(TRUE). Priors were specified as weakly informative normal(0, 1) distributions on the population-level slopes (class b) and exponential(1) distributions on the group-level standard deviations (class sd) and residual standard deviations (class sigma) for both response variables.

POMS2-related analyses

Associations involving owner mood (POMS2) were examined at the level of individual owner–cat dyads. Because owner mood was assessed once per owner, its values are shared across cats belonging to the same household; we therefore first attempted to account for this non-independence by including owner identity as a group-level (random) intercept. However, because owner mood does not vary within a household, the group-level intercept absorbed all between-owner variance, and these models failed to converge ($\widehat{R} > 3$). We therefore report dyad-level models without a group-level term as the primary analysis, and additionally conducted sensitivity analyses in which all variables were aggregated to the household level (mean across cats within a household; $n = 19$ households for owner hormone and relationship variables, $n = 17$ for cat hormone variables). The direction and statistical support of the primary associations were largely consistent between the two approaches, with the exception of the association between cohabitation duration and owner mood, which was not supported in the household-level sensitivity analysis. Full results of both the dyad-level and household-level models are provided in Tables S3–S5 and S6

All variables were standardized (z-scored) prior to analysis; POMS2 subscale scores were standardized within subscale. Models were fitted using brms with its default weakly informative priors: a flat (improper uniform) prior on the population-level slope (class b), a Student's t(3, ν, 2.5) prior on the intercept—where ν was automatically centered by brms on the empirical location of each standardized outcome and therefore varied slightly across models—and a Student's t(3, 0, 2.5) prior (truncated at zero) on the residual standard deviation (sigma).

Association between human and cat hormones and owner mood

Associations between hormone concentrations and owner mood were examined using separate Bayesian Gaussian regression models for each of the seven POMS2 subscales and Total Mood Disturbance and each of the four hormone measures (owner oxytocin, owner cortisol, cat oxytocin, cat cortisol), yielding 32 univariate models ($n = 29$ dyads for owner oxytocin and owner cortisol; $n = 24$ dyads for cat oxytocin and cat cortisol, reflecting five cats for which urine could not be collected; see Method Details, Urine sample collection).

Association between proximity, cohabitation duration, and owner mood

Associations between proximity (median beacon RSSI during owner presence), cohabitation duration, and owner mood were examined using separate Bayesian Gaussian regression models for each of the seven POMS2 subscales and Total Mood Disturbance, with proximity and cohabitation duration entered as single predictors (16 univariate models total; $n = 29$ dyads).

Secure Base Test (SBT) models

To examine associations between hormonal measures and approach behavior in the Secure Base Test (SBT), the duration of time cats spent inside the designated circle area was used as the primary behavioral index. Because the behavioral data included excess

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zeros, Bayesian hurdle Gamma regression models were fitted separately for cat oxytocin and cat cortisol concentrations. In each model, target individual type (owner vs. stranger), phase, and standardized hormone concentrations were entered as predictors together with all interaction terms. Cat identity, nested within owner identity, was included as a group-level (random) effect, to account for both repeated observations within cats and potential non-independence among cats from the same household. The hurdle component modeled the probability of whether cats entered the circle area at all (zero vs. non-zero responses), whereas the conditional Gamma component modeled the duration spent inside the circle among non-zero observations.

Models used brms' default weakly informative priors: flat (improper uniform) priors on the population-level slopes for both the conditional Gamma component (class b) and the hurdle component (class b, dpar = hu); a Student's t(3, −2.3, 2.5) prior on the Gamma component intercept, automatically centered on the empirical location of the (log-link) outcome; a standard logistic(0, 1) prior on the hurdle component intercept; Student's t(3, 0, 2.5) priors (truncated at zero) on all group-level standard deviations (owner identity, and cat identity nested within owner identity, for both components); and a weakly informative gamma(0.01, 0.01) prior on the shape parameter of the Gamma distribution.

Software and MCMC settings

Unless otherwise specified, all models were fitted using the brms package in R with 4 Markov chain Monte Carlo chains of 6,000 iterations each (3,000 warmup, 3,000 post-warmup draws per chain; 12,000 post-warmup draws total), and convergence was assessed via $\widehat{R}$ (≤ 1.01) and the absence of divergent transitions.